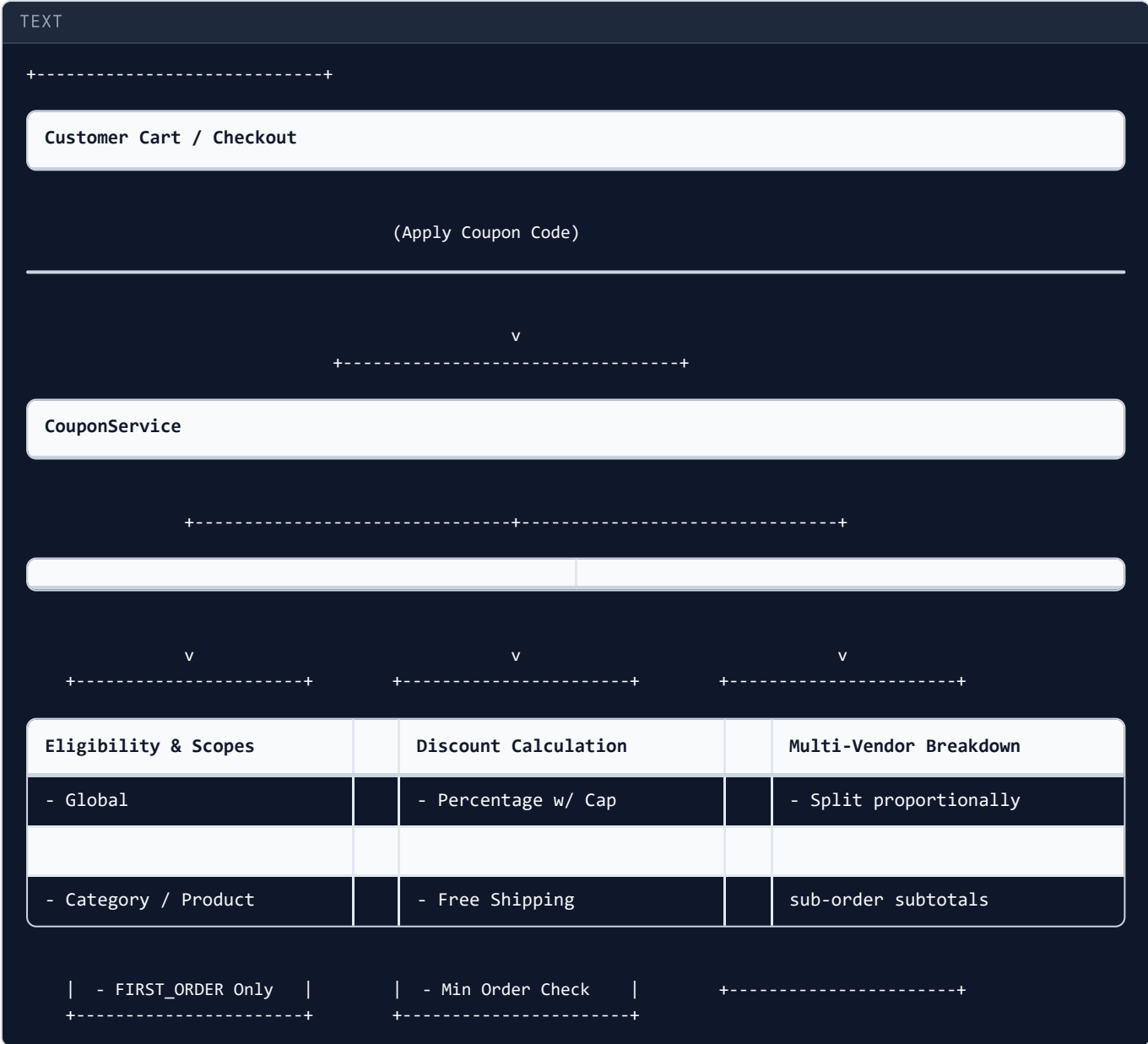


Stage 16: Coupons, Promotions, Flash Deals & Referral Engine

1. Architectural Overview

The **Coupons, Promotions, Flash Deals & Referral Engine** provides an enterprise-grade discount calculation and promotional engine for the Alight International Multi-Vendor Marketplace. It ensures exact arithmetic precision, granular scoping (Global, Vendor, Category, Product, First Order), fair multi-vendor proportional discount splitting, usage quota enforcement, and time-bounded promotional flash campaigns.



2. Core Entities & Database Schema (`v42__coupons_promotions_and_flash_deals.sql`)

2.1 Coupons (`coupons`)

- `code` : Unique uppercase code index (e.g., `SUMMER20` , `WELCOME100`).
- `discount_type` : `PERCENTAGE` , `FIXED_AMOUNT` , `FREE_SHIPPING` .
- `discount_value` : Numerical value of discount (percentage rate or fixed amount).
- `max_discount_amount` : Upper ceiling cap for percentage discounts.
- `min_order_amount` : Minimum eligible cart subtotal threshold required.
- `scope` : `GLOBAL` , `VENDOR` , `CATEGORY` , `PRODUCT` , `FIRST_ORDER` .
- `usage_limit_total` & `usage_limit_per_user` : Global and per-account redemption limits.
- `referral_user_id` : Associated user for customer referral discount attribution.
- `terms_and_conditions` : Legal and policy disclosure text.

2.2 Coupon Usages (`coupon_usages`)

- Tracks per-order coupon redemptions (`coupon_id` , `user_id` , `order_id` , `discount_amount` , `used_at`).

2.3 Promotions & Flash Deals (`promotions`)

- `banner_tag` : Tag identifier (e.g., `HERO_BANNER` , `FLASH_DEAL` , `FESTIVAL_OFFER`).
- `badge_text` & `discount_text` : Visual pill labels and copy for storefront carousels.
- `start_time` & `end_time` : ISO-8601 validity windows for automated deal expiration.
- `display_order` : Priority sequence for homepage and category banner sorting.

3. Discount Allocation & Multi-Vendor Mechanics

When a coupon is applied to a multi-vendor cart:

1. **Scope Filtering:** Items are evaluated against the coupon's scope.

- For `GLOBAL` or `FIRST_ORDER` , all items are eligible.
- For `VENDOR` , only items matching the coupon's vendor are eligible.
- For `CATEGORY` / `PRODUCT` , only matching catalog items are eligible.

2. **Minimum Order Threshold:** The sum of eligible item line totals must meet `min_order_amount` .

3. **Discount Computation:**

- **Percentage:** `(eligibleSubtotal * discountValue / 100).min(maxDiscountAmount)` .
- **Fixed Amount:** `discountValue.min(eligibleSubtotal)` .
- **Free Shipping:** Sets shipping total to 0.

4. **Multi-Vendor Proportional Splitting:**

- For each eligible vendor sub-order, discount is split as:

$$\$ \text{Vendor Discount} = \text{Total Discount} \times \frac{\text{Vendor Eligible Subtotal}}{\text{Total Eligible Subtotal}} \$$$

- Guarantees exact reconciliation on vendor sub-orders, payout calculations, and escrow deposits.

4. Verification & Integration Suite

Verified via `CouponsAndPromotionsIntegrationTest.java` :

- ✅ **Global Percentage Discount:** Max cap enforcement.
- ✅ **Fixed Amount Discount:** Minimum order threshold validation.
- ✅ **Free Shipping Coupon:** Shipping fee waiver calculation.
- ✅ **Multi-Vendor Scoped Coupon:** Proportional vendor discount breakdown.
- ✅ **First-Time Buyer Restriction:** `FIRST_ORDER` scope checking.
- ✅ **Redemption Limits:** Per-user and global quota enforcement via `coupon_usages` .
- ✅ **State Transitions:** Active/Inactive toggling and expiration date validation.
- ✅ **Admin & Vendor CRUD:** Full administrative lifecycle and stats queries.
- ✅ **Promotion Banners & Flash Deals:** Scheduling and display ordering.
- ✅ **End-to-End Checkout:** Coupon deduction on checkout and order creation.